

HazardMind AI Executive Disaster Report

Event ID	demo-peshawar-flood
Location	Peshawar, Pakistan
Hazard Type	Flood
Overall Severity	CRITICAL
Satellite Type	sentinel-1
Satellite Reason	cloud_cover_above_30_percent_sar_selected

Operational Statistics

Affected Area	153.37 km2
Damage Percent	24.3%
Total Zones	22
Population Affected	540,000
Hospitals at Risk	14
Roads Blocked	89 km
Schools Affected	67
Vulnerability Score	8.2

Generated Risk Map

- 153.37 km² total inundated area across 22 zones
- Severe infrastructure impact: 14 hospitals and 67 schools affected
- Logistical bottleneck: 89 km of blocked roads with only one active evacuation route
- High vulnerability score of 8.2 across affected residential and commercial sectors

Priority Timeline

Next 6 Hours	Evacuate high-risk zones FZ-01 and FZ-02 immediately; Secure Lady Reading Hospital and deploy medical teams to affected zones.
Next 24 Hours	Clear primary road corridors to restore rescue access; Open and staff temporary shelters for 540,000 affected residents.
Next 72 Hours	Restore access to 67 affected schools and public buildings; Implement long-term drainage and water removal plans.
Resources	High-clearance rescue vehicles and boats; Emergency medical supplies and surge staffing; Heavy machinery for road clearing.
Coordination	Sync satellite data with local ground-truth reports; Coordinate multi-agency traffic management for evacuation routes.

Anomalies and Warnings

- MEDIUM: PDF and Map URLs are empty Handling: Verify report generation pipeline and attach assets
- GeoJSON artifact URL is not available in local demo mode.
- Map URL is generated later by the local artifact step.
- Data inconsistency: routes.evacuation_route_count is 0, but narrative reports one operational route remaining.

Quality Check

Status	not_ready
Event Id Present	True
Satellite Data Present	True
Hazard Data Present	True
Impact Data Present	True
Map Artifacts Present	True
Recommendations Present	True
Confidence Above Threshold	True
Warnings	GeoJSON artifact URL is not available in local demo mode.; Map URL is generated later by the local artifact step.
Blocking Issues	Missing critical deliverables: pdf_url and map_url are empty.

Band-Ready Final Message

HazardMind Report Agent complete for demo-peshawar-flood (Peshawar, Pakistan, Flood). Criticality: critical; confidence: 92%. Summary: A CRITICAL flood event in Peshawar has inundated 153.4 km² across 22 mapped zones, affecting 540,000 residents and blocking major evacuation routes.

zones, affecting approximately 540,000 people with 91% SAR confidence. Infrastructure impact is severe, with 14 hospitals at risk, 89 km of roads blocked, and 67 schools affected. The most urgent risk is that only one evacuation route remains operational, constraining egress from critical zones FZ-01 and FZ-02. Priorities for the next 6 hours are immediate evacuation of FZ-01/FZ-02, securing Lady Reading Hospital, and establishing traffic control on the remaining route. All figures are satellite-derived estimates requiring ground-truth validation.

Detailed Incident Analysis

A critical flood event in Peshawar, Pakistan, has affected 153.37 km², impacting 540,000 people. SAR VV/VH analysis confirms a mean ratio of 0.24, indicating widespread inundation with 24.3% of the area damaged. The disaster map comprises 22 zones, including a critical deep-water zone (FZ-01, 12.4 km²) and a high-severity water zone (FZ-02, 8.7 km²). Critical infrastructure is threatened, with 14 hospitals at risk, 89 km of roads blocked, and 67 schools affected. Only one evacuation route remains available, heightening the emergency.

Technical Analysis

Sentinel-1 SAR was selected because cloud cover is 42%. The analysis uses SAR VV/VH ratio with mean value 0.24 and identifies 22 zones. Damage is estimated at 24.3%, with 89 km of roads blocked and a vulnerability score of 8.2.

Recommendations

- Prioritize evacuation in critical flood zones.
- Deploy emergency medical support near hospitals at risk.
- Clear blocked road corridors for rescue access.
- Open temporary shelters for displaced families.
- Monitor flood expansion using updated satellite imagery.
- Evacuate high-risk zones FZ-01 and FZ-02 immediately
- Secure Lady Reading Hospital and deploy medical surge teams
- Establish emergency traffic control on the remaining evacuation route
- High-clearance rescue vehicles and boats
- Emergency medical supplies and surge staffing

Response Priorities

- Activate evacuation support for FZ-01 and FZ-02.
- Protect hospital access and stage medical surge teams near high-risk facilities.
- Clear blocked road corridors needed for rescue and logistics movement.

Assumptions

- Satellite, hazard, and impact values are local demo data until upstream agents publish live outputs.
- Risk zone geometries are simplified mock polygons for dashboard and report demonstration.
- Facility and route coordinates are approximate and intended for prototype visualization.

Limitations

- No live Band messages are consumed in this local demo run.
- No field validation, hydrologic model, or live basemap tiles are used for artifact generation.
- Impact estimates should be treated as decision-support indicators, not official counts.

Model Source Note

Detailed report generated using Featherless Kimi K2.6. Executive summary generated using AI/ML API. Intelligence sources: criticality: featherless:kimi-k2.6, anomaly_check: featherless:gemma-4-31b-it, map_narrative: featherless:gemma-4-31b-it, priority_recommendations: featherless:gemma-4-31b-it, decision_brief: aiml:opus-4.8, quality_check: hybrid:deterministic+featherless:gemma-4-31b-it, band_ready_message: template+intelligence

Detailed Report	featherless:composite
Executive Summary	aiml:opus-4.8
Fallback Used	False
Featherless Model	moonshotai/Kimi-K2.6
Criticality	featherless:kimi-k2.6
Map Narrative	featherless:gemma-4-31b-it
Priority Timeline	featherless:gemma-4-31b-it
Quality Check	hybrid:deterministic+featherless:gemma-4-31b-it

Agent Log / Trace

Agent	Status	Timestamp	Message
hazardmind-report	complete	2026-06-13T18:03:00Z	Report Agent received event context
hazardmind-report	complete	2026-06-13T18:03:20Z	Featherless composite generated detailed disaster report
hazardmind-report	complete	2026-06-13T18:03:40Z	AI/ML generated executive summary
hazardmind-report	complete	2026-06-13T18:03:45Z	Criticality assessed
hazardmind-report	complete	2026-06-13T18:03:50Z	Anomalies checked
hazardmind-report	complete	2026-06-13T18:03:55Z	Map narrative generated
hazardmind-report	complete	2026-06-13T18:04:00Z	Priority timeline generated
hazardmind-report	complete	2026-06-13T18:04:05Z	Decision brief generated with Opus
hazardmind-report	complete	2026-06-13T18:04:10Z	Quality check completed
hazardmind-report	complete	2026-06-13T18:04:15Z	Band-ready final message prepared
hazardmind-report	complete	2026-06-13T18:04:00Z	Static cartography map generated locally